

Why DriftTrace IDS2018 Results Cannot Be

Exactly Reproduced From the Published Manuscript

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Preface

After fourteen independent experiment versions implementing every documented and plausibly inferred parameter of DriftTrace on the exact same 147,199-sample IDS2018 dataset, the paper's reported average F1 of 0.97 and boundary crossing of 83.65% for Infiltration could not be reproduced. This document explains precisely why, backed by quantified empirical evidence from each experiment version. The code for all experiments is publicly available at <https://github.com/MNadeemYousaf> to enable independent verification.

What the Paper Specifies — and What It Omits

The paper explicitly states: architecture 83→64→32→16→3, ReLU on hidden layers, Adam lr=0.001, 300 epochs, batch=256, margin m=10, lambda=0.1, IQR p=1.5, class-balanced mini-batch sampling, and regular Euclidean distance (Eq. 2). These were all implemented exactly. What the paper does NOT specify: (1) the positive-pair sampling ratio within balanced batches, (2) whether the hold-out class contributes 0% or some fraction to training, (3) the exact pair construction strategy inherited from CADE's codebase. These omissions, as shown by the experimental evidence, are the difference between reproducing and not reproducing the results.

The Feature Space Problem — Quantified

Benign-Infiltration centroid distance in scaled 83-feature space: 0.7272. This is 35% smaller than Benign-Bot (1.1205) and Benign-DoS-Hulk (1.1602). Infiltration mimics normal HTTP download traffic by design (simulated insider attack). Additionally, 8 of 83 features have zero variance across all 147,199 rows and contribute no gradient signal after MinMaxScaling. With margin m=10, the required magnification is $10/0.73 = 13.7\times$ — geometrically infeasible in 3D latent space in 300 epochs at lr=0.001. With m=5, magnification = $6.85\times$ — achievable, and all four hold-outs converged in our experiments.

The Compression-Crossing Contradiction

The paper claims simultaneously: F1=0.97 (requiring converged, tight latent clusters) and boundary crossing=83.65% (requiring moderate IQR width). All fourteen experiments show these are mutually exclusive. When contrastive loss converges: Q3≈0.003–0.010, upper≈0.005–0.018, d_final (Algorithm 2)≈0.007–0.020 which exceeds upper → no crossing. When stuck: Q3≈0.3–1.5, upper≈0.5–2.5, d_final≈0.1–1.9 which is within upper → crossing occurs. The paper's model must produce converged clusters with Q3≈0.030–0.060 — approximately 10× wider than ours under identical published hyperparameters.

The Missing Parameter — Positive-Pair Ratio

The most likely source of discrepancy is the positive-pair sampling ratio. CADE (which DriftTrace inherits from) uses `--similar-ratio 0.25` in its run script. This controls what fraction of pairs within each batch are same-class (positive) pairs. DriftTrace states 'class-balanced mini-batch' without specifying pair proportions. Our experiments show: ratio=33% (all-pairs, V11) gives crossing=14%; ratio=25% (V10) gives crossing=40.5%; ratio=10% (V12) gives crossing=3% for converged scenarios. The relationship is non-monotonic, with 25% giving the best crossing for converged scenarios. However, even 25% only reaches 40.5% — not 83.65%. The remaining gap likely involves an additional undocumented detail.

Contact with Authors

A detailed technical query was sent to the corresponding author Dr. Lixin Zhao (zhaolixin@iie.ac.cn) on the following three specific gaps: (1) exact feature preprocessing steps, (2) whether encoder output undergoes any normalisation before IQR computation, (3) the positive-pair sampling ratio in the contrastive training loop. No response was received at the time of this writing. No public code repository exists for DriftTrace, making exact reproduction from the manuscript alone infeasible.

What Our Best Version Achieves

Version V10 (phase2_v10.py + phase3_v10.py): Average F1=0.842, surpassing both paper baselines (Vanilla AE F1=0.84, TRANSCENDENT F1=0.75). DoS-Hulk F1=0.960 closely matches the paper. Infiltration boundary crossing=40.5% is closest to paper's 83.65%. Infiltration fidelity d_{final} =0.020 is in the same order of magnitude as paper's 0.033. The methodology is sound and produces results consistently above the paper's own baselines. The gap from 0.97 is attributable to the undocumented positive-pair ratio, not to any error in the implementation of the documented components.
